

Research Interests

- Atmospheres of brown dwarfs and (exo)planets
- Machine learning applications in astrophysics

Education

- 2024 – **PhD in Astronomy**, *University of Hertfordshire, UK*
– **Project:** *Remote Sensing Alien Clouds and Chemistry with JWST*
– **Supervisors:** Dr. Ben Burningham, Dr. Fei Wang, Prof. Hugh Jones
- 2021 – 2023 **MSc in Astronomy Research**, *Leiden University, Netherlands*, Grade: 7.5/10 (US: A-, UK: Upper Second Class)
– **2nd MSc Thesis:** *Impact of Thermal Evolution on Exoplanetary Atmospheres*
– **Supervisors:** Dr. Yamila Miguel, Amy Louca

– **1st MSc Thesis:** *Substructure of Planet-Forming Disks with VLT*
– **Supervisors:** Prof. Dr. Michiel Hogerheijde, Dr. József Varga
- 2020 – 2021 **MSc in Astronomy**, *Eötvös Loránd University, Hungary*, Discontinued, went to Leiden instead
- 2017 – 2020 **BSc in Earth Sciences**, *Eötvös Loránd University, Hungary*, Grade: 5.0/5.0 (US: A+, UK: First Class)
– **BSc Thesis:** *Lightcurves of Hilda Asteroids and Transneptunian Objects*
– **Supervisor:** Dr. Csaba Kiss

Work Experience

- Nov 2024 – **Visiting Lecturer**, *Department of Physics, Astronomy and Maths, University of Hertfordshire, UK*
Present ○ Leading PC laboratories and grading assignments for Data Science MSc students.
 - Machine Learning and Neural Network: Fall 2025, Spring 2026
 - Applied Data Science 1: Spring 2026
 - Data Handling and Visualisation: Spring 2025
- Sep 2022 – **Teaching Assistant**, *Leiden University, Faculty of Science, Netherlands*
Dec 2022 ○ Led practical sessions and graded weekly assignments for the mandatory Astronomy BSc course, 'Stars'.
- Oct 2019 – **Observatory Assistant**, *Konkoly Observatory, HUN-REN Research Centre for Astronomy and Earth Sciences, Hungary*
Aug 2021 ○ Performed observations at Piszkestető Mountain Station (3-4 nights every month).
○ Assisted in data reduction and analysis of the observed data.

Publications

First-authored

4. **Kecskeméthy, V.**; Burningham B., et al. (2026) *A Neural Network Forward Model for High-Dimensional Retrievals of Brown Dwarf Atmospheres*. (In preparation).
3. **Kecskeméthy, V.**; Burningham B., et al. (2026) *Benchmark Brown Dwarfs as Chemical Laboratories: Linking System Bulk Properties to Atmospheric Retrievals*. (Accepted for publication in MNRAS). [\[arXiv\]](#)
2. **Kecskeméthy, V.**; Louca, A.; Miguel, Y. (2024). *Temporal Variability in Transmission Spectra of H₂-Dominated Exoplanets: The Influence of Thermal Evolution and Stellar Irradiation on Atmospheric Composition*. *New Astronomy*, Volume 113, id.102296, December 2024. [\[ADS\]](#)
1. **Kecskeméthy, V.**; Kiss, C., et al. (2023). *Light Curves of Transneptunian Objects from the K2 Mission of the Kepler Space Telescope*. *The Astrophysical Journal Supplement Series*, Volume 264, Issue 1, id.18, 20 pp., January 2023. [\[ADS\]](#)

Co-authored

5. Rothermich, A.; [and 18 others, including **Kecskeméthy, V.**] (2026). *The Buddy System: Characterizing the benchmark system containing the T0 brown dwarf CWISE J210640.16+250729.0 using JWST*. Submitted to AAS
4. Varga, J.; [and 39 others, including **Kecskeméthy, V.**] (2024). *Mid-infrared Evidence for Iron-Rich Dust in the Multi-Ringed Inner Disk of HD 144432*. *Astronomy & Astrophysics*, Volume 681, id.A47, 29 pp., January 2024. [\[ADS\]](#)

3. Csörnyei, G.; [and 9 others, including **Kecskeméthy, V.**] (2022). *Study of Changes in the Pulsation Period of 148 Galactic Cepheid Variables*. *Monthly Notices of the Royal Astronomical Society*, Volume 511, Issue 2, pp.2125-2146, April 2022. [\[ADS\]](#)
2. Marciniak, A.; [and 67 others, including **Kecskeméthy, V.**] (2021). *Properties of Slowly Rotating Asteroids from Convex Inversion Thermophysical Model*. *Astronomy & Astrophysics*, Volume 654, id.A87, 32 pp., October 2021. [\[ADS\]](#)
1. Kalup, C.; [and 10 others, including **Kecskeméthy, V.**] (2021). *101 Trojans: A Tale of Period Bimodality, Binaries, and Extremely Slow Rotators from K2 Photometry*. *The Astrophysical Journal Supplement Series*, Volume 254, Issue 1, id.7, 20 pp., May 2021. [\[ADS\]](#)

Talks & Posters

5. **Benchmark Brown Dwarfs as Chemical Laboratories** (Talk) Santa Cruz, CA, USA
Other Worlds Laboratory (OWL) 2026
4. **Silicate Clouds and Chemical Inheritance in Benchmark Brown Dwarfs** (Talk) Monterey, CA, USA
Charting Substellar Worlds 2026
3. **Investigating Silicate Clouds in Benchmark L-T Dwarfs with JWST** ([Poster](#)) Cork, Ireland
European Astronomical Society Annual Meeting 2025
2. **SDSSJ141659.78+500626.4: Investigating Silicate Clouds on an L dwarf** (Talk) Hatfield, UK
UH-PAM Lunch talk 2025
1. **NH₃ observations of the Orion Molecular Cloud** (Talk) Budapest, Hungary
Interaction of Stars with their Environment 2019

Scholarships, Awards & Grants

- Aug 2026 **Other Worlds Laboratory (OWL) Travel Grant**, UC Santa Cruz
- July 2026 **Charting Substellar Worlds Travel Grant**, UC Santa Cruz
- 2024 – 2028 **Science and Technology Facilities Council (STFC) PhD Studentship**, University of Hertfordshire
- Spring 2020 **Research Scholarship (tudományos ösztöndíj)**, Eötvös Loránd University
- Dec 2019 **Young Scientists' Conference (TDK)**: Submitted 2 projects, awarded 1st and 2nd place, Eötvös Loránd University
- Spring 2018 – **Academic Excellence Scholarship (tanulmányi ösztöndíj)**: Won it four times, Eötvös Loránd University
- Fall 2019